Multi-Method Characterization of Silicon-on-Insulator Interconnects: Cross-Validating Waveguide Dispersion and Loss via Linear and Resonant Interferometry

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***Abstract*—This proposal details the design, simulation, and planned experimental validation of silicon-on-insulator (SOI) photonic integrated circuits. The primary objective is the high-precision extraction of the waveguide group index (n_g) and propagation loss through a hybrid characterization approach utilizing unbalanced Mach-Zehnder Interferometers (MZI) and add-drop ring resonators. By enabling systematic assessment of fabrication-induced variability, we cross-validate Finite Difference Eigenmode (FDE) solvers with circuit-level simulations for 500×220 nm strip waveguides. The proposed chip layout is optimized for the SiEPIC E-Beam fabrication process.**

I. Introduction

Silicon-on-insulator photonic integrated circuits enable high-index-contrast light confinement, supporting compact interferometric and resonant architectures. However, the high modal confinement in SOI waveguides renders their phase response significantly sensitive to nanometer-scale geometric variations [1]. This sensitivity results in waveguide dispersion, necessitating accurate characterization of the effective index (n_{eff}), group index (n_g), and propagation loss.

Relying on a single device architecture often introduces systematic measurement bias. Consequently, this research utilizes both linear MZIs and resonant ring structures. As established in recent literature [3], agreement between n_g values extracted from both configurations provides independent verification of the dispersion model and isolates systematic geometric errors from coupling-induced spectral shifts.

II. Waveguide Theory and Modal Analysis

A. FDE Solver and Light Confinement

Our standard waveguide is a silicon strip that is 500 nm wide and 220 nm tall, sitting on a $2\mu m$ layer of silicon dioxide (BOX). We first used a Finite Difference Eigenmode (FDE) solver to see how light behaves inside this shape. As shown in Fig. 1, the fundamental TE mode is tightly confined inside the silicon core.

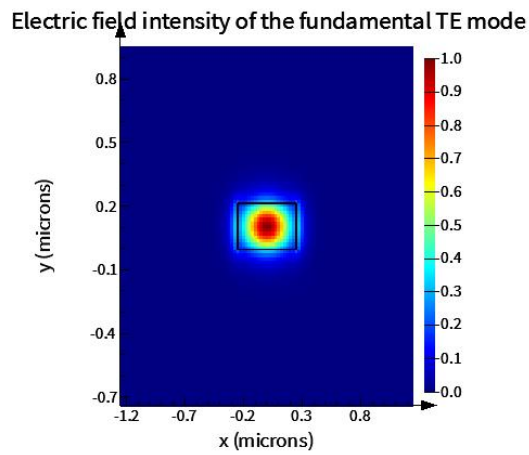

Fig. 1: Normalized electric field intensity ($|E|^2$) of the fundamental TE mode at $\lambda = 1550$ nm.

B. Dispersion and Group Index (n_g) Extraction

The effective index (n_{eff}) tells us the phase velocity of the light, but because this index changes depending on the wavelength, a pulse of light actually travels at the group index (n_g):

$$n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \tag{1}$$

Numerical analysis yields $n_g \approx 4.21$ at 1550 nm (Fig. 2).

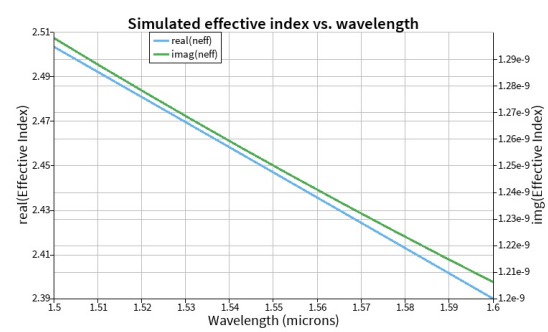

Fig. 2: Wavelength-dependent effective index (n_{eff}) simulation results.

C. Compact Model for Circuit Simulation

We fit the n_{eff} data to a second-order polynomial centered at $\lambda_0 = 1550$ nm:

$$n_{eff}(\lambda) \approx n_0 + n_1(\lambda - \lambda_0) + n_2(\lambda - \lambda_0)^2 \tag{2}$$

Coefficients: $n_0 = 2.446820$, $n_1 = -1.133390\mu m^{-1}$, and $n_2 = -0.043937\mu m^{-2}$.

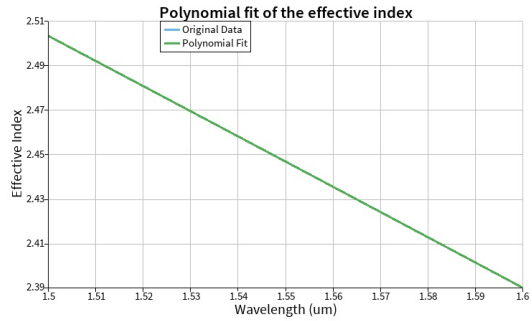

Fig. 3: Verification of the polynomial compact model fit.

III. Design of Interferometric Structures

A. Mach-Zehnder Interferometer (MZI)

An MZI creates a phase shift ($\Delta\phi$) via a length difference ΔL:

$$\Delta\phi = \frac{2\pi n_{eff}\Delta L}{\lambda} \tag{3}$$

The total power transmission $T(\lambda)$ follows a cosine wave pattern:

$$T(\lambda) = \frac{1}{2}\left[1 + \cos\left(\frac{2\pi n_{eff}\Delta L}{\lambda}\right)\right] \tag{4}$$

The Free Spectral Range (FSR) is used to calculate the real group index:

$$FSR_{MZI} = \frac{\lambda^2}{n_g \Delta L} \tag{5}$$

B. Ring Resonators

Resonance occurs when $m\lambda = n_{eff} 2\pi R$. The FSR for the ring is:

$$FSR_{ring} = \frac{\lambda^2}{n_g (2\pi R)} \tag{6}$$

C. Measurement Constraints and Loss Logic

The selected ΔL and radius values ensure sufficient sampling density within the grating coupler bandwidth (50 nm). The propagation loss α is extracted from the Quality Factor ($Q = \lambda_{res}/FWHM$). To achieve high precision, resonance dips will be treated using a Lorentzian fit:

$$T(\lambda) = 1 - \frac{A}{1 + \left(\frac{\lambda - \lambda_{res}}{FWHM/2}\right)^2} \tag{7}$$

Mathematically, Q relates to the total cavity loss (intrinsic and coupling) via [2]:

$$Q_{total}^{-1} = Q_{intrinsic}^{-1} + Q_{coupling}^{-1} \tag{8}$$

By sweeping the gap variable g (15, 20), we modify the coupling coefficient κ. The point of maximum Extinction Ratio (ER) indicates Critical Coupling, allowing for the decoupling of α from external coupling losses.

IV. Results and Layout Presentation

A. Predicted FSR and GDSII Layout

The tables below summarize the parameters for the MZI devices and the Ring configurations.

TABLE I: Complete MZI Parameter Sweep (ΔL)

Device ID	Path Imbalance (ΔL)	Expected FSR (nm)
MZI 1	25 μm	22.82
MZI 2	50 μm	11.41
MZI 3	100 μm	5.70
MZI 4	150 μm	3.80
MZI 5	200 μm	2.85

TABLE II: Complete Ring Resonator Array (R and g)

Ring ID	Radius (R)	Gap (g)	Expected FSR (nm)
1, 2	5 μm	15, 20	18.20
3, 4	6 μm	15, 20	15.17
5, 6	7 μm	15, 20	13.00
7, 8	8 μm	15, 20	11.38
9, 10	9 μm	15, 20	10.11

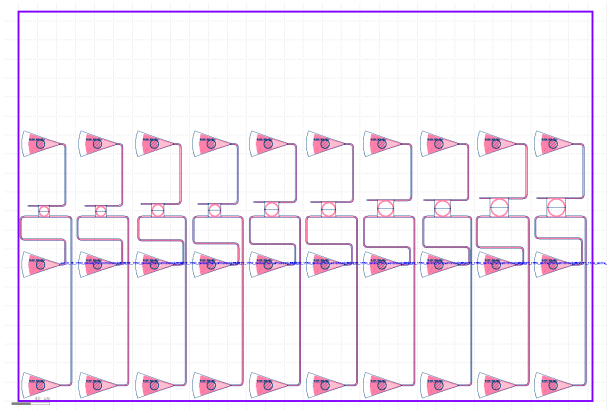

Fig. 4: Comprehensive GDSII layout showing the integrated Ring Resonator array.

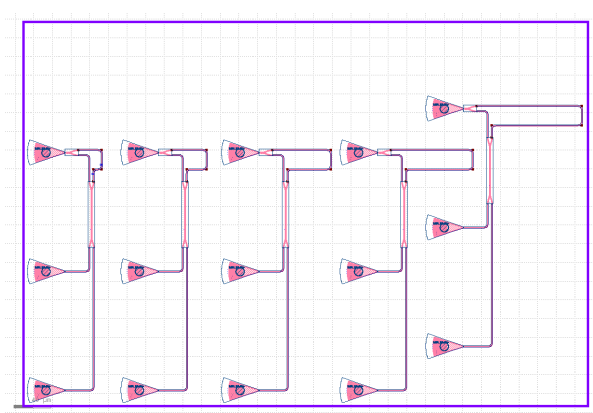

Fig. 5: GDSII layout detail of the MZI test structures illustrating the arm imbalance (ΔL).

B. Spectral Verification

Simulations of the $R = 5\mu m$ ring (Fig. 6) and the $\Delta L = 50\mu m$ MZI (Fig. 7) verify the predicted phase and resonant behavior.

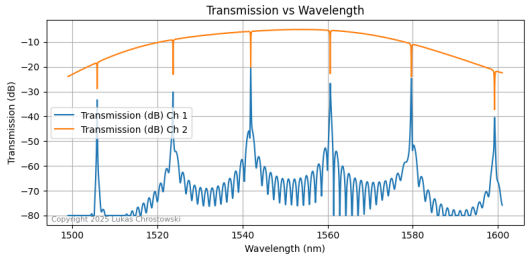

Fig. 6: Simulated Add-Drop response of the $5\mu m$ ring resonator.

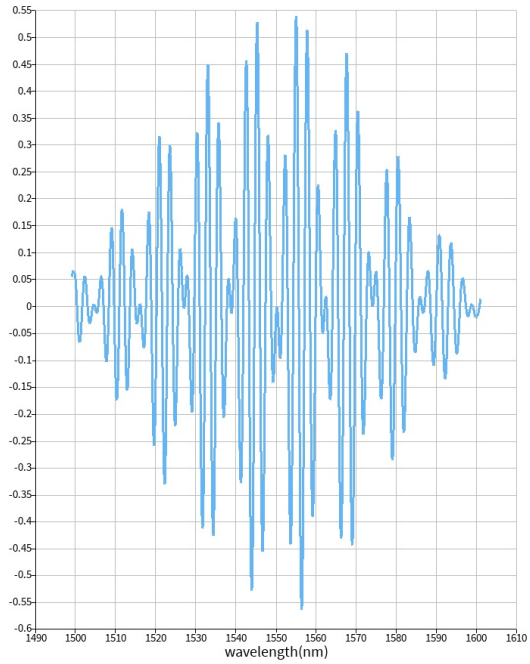

Fig. 7: Preliminary MZI spectral response for phase verification.

V. Conclusion

This proposal provides a self-consistent method for extracting both the group index and propagation loss by integrating multiple interferometric architectures on a single chip.

References

[1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design*. Cambridge, 2015.

[2] W. Bogaerts et al., "Silicon microring resonators," *Laser & Phot. Rev.*, 2012.

[3] B. Jalali and S. Fathpour, "Silicon Photonics," *J. Lightwave Technol.*, 2006.